Fiscal Year:	FY 2005	**Task Last Updated:**	FY 08/13/2005
PI Name:	Merfeld, Dan Ph.D.		
Project Title:	Influence of Sensory Integration on the Neural Processing of Gravito-Inertial Cues		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	Dan_Merfeld@meei.harvard.edu	**Fax:**	FY (617) 573-5596
PI Organization Type:	UNIVERSITY	**Phone:**	(617) 573-5595
Organization Name:	Massachusetts Eye and Ear Infirmary		
PI Address 1:	Jenks Vestibular Physiology Laboratory		
PI Address 2:	243 Charles Street		
PI Web Page:			
City:	Boston	**State:**	MA
Zip Code:	02114	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	99-HEDS-02
Start Date:	12/01/2003	**End Date:**	09/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	2
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Lionel, Zupan Ph.D. (MEEI)		
Grant/Contract No.:	NNJ04HB01G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Our sense of spatial orientation results from a complex set of neural processes of sensory integration; these processes utilize information from many different physiological systems. Evidence suggests that extended exposure to micro-gravity yields adaptive changes in these neural processes. For the most part, these adaptive changes appear functionally relevant for spaceflight but yield significant transient problems when astronauts return to gravitational environments (Earth, Mars, Moon, etc.). In-flight countermeasures that reinforce the normal sensory interactions encountered in a gravitational environment (e.g., rotations about any axis not aligned with gravity are accompanied by changes in the relative orientation of gravity) will almost certainly help astronauts function adequately when they transition to a gravitational environment. This will be important to pilot astronauts landing spacecraft as well as to crewmembers trying to rapidly exit a spacecraft. Furthermore, artificial gravity has been proposed as a countermeasure for numerous problems associated with spaceflight (including cardiovascular deconditioning, bone decalcification, etc.).		

	One undesired side effect of artificial gravity will be additional alterations in these processes of sensory integration. For all of these reasons, it is essential that we understand the predominant sensory interactions that are altered by exposure to micro-gravity. Specifically, we proposed to investigate how rotational signals from the semicircular canals influence how humans process the ambiguous gravito-inertial force measured by the otolith 0rgans. We are now performing a series of ground experiments that seek a better understanding of sensory interactions between the semicircular canals and graviceptors. We propose to perform these investigations using several different measurement techniques including measures of reflexive eye responses as well as psychophysical measures of human perception.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	In our studies, we are investigating how normal subjects process ambiguous sensory cues from the otolith organs, which measure gravity and linear acceleration. These basic science studies will lead to improved diagnostic testing for human patients suffering from balance and vestibular problems.
Task Progress:	In the past year, we have published two papers. We have also had a 3rd manuscript accepted for publication. We are presently working on 2 additional manuscripts that we plan to submit in the coming year. We have also installed and tested a new motion device that is capable of simultaneously tilting and translating human subjects. This "Tilt Device" is nearly operational. We anticipate that it will be available for human testing before the end of this month. In order to accomodate this device and 2 other new devices, our institution funded the development of 1200 square feet of new, modern, laboratory space. The construction of this space has been completed and we have begun occupying this new research space. Since this space was essential in order for us to pursue the funded research, this was a very important step of progress toward our goals of completing the proposed research.
Bibliography Type:	Description: (Last Updated: 04/05/2011)
Articles in Peer-reviewed Journals	Merfeld DM, Park S, Gianna-Poulin C, Black FO, Wood S. "Vestibular perception and action employ qualitatively different mechanisms. I. Frequency response of VOR and perceptual responses during Translation and Tilt." J Neurophysiol. 2005 Jul;94(1):186-98. PMID: 15728767 , Jul-2005
Articles in Peer-reviewed Journals	Merfeld DM, Park S, Gianna-Poulin C, Black FO, Wood S. "Vestibular perception and action employ qualitatively different mechanisms. II. VOR and perceptual responses during combined Tilt&Translation." J Neurophysiol. 2005 Jul;94(1):199-205. PMID: 15730979 , Jul-2005